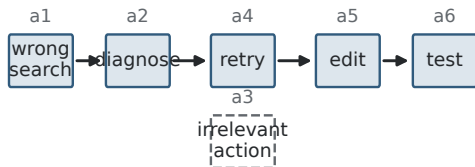


(a) Declared six-action precedence graph

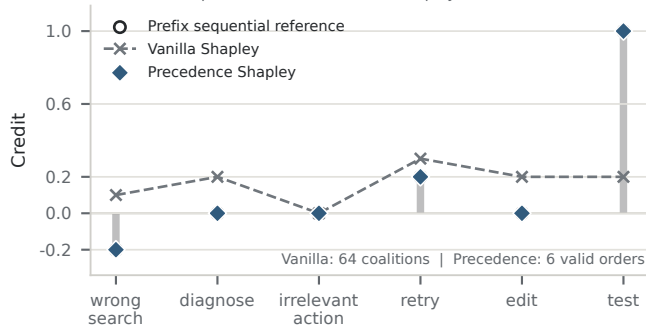
Representative protocol: failure_recovery-00



71.875% mean impossible-coalition rate under unconstrained counterfactual subsets (40 cases)

(b) Action credit on a recovery trajectory

Declared prefix reference vs. exact Shapley variants



(c) Aggregate estimator comparison

40 controlled cases across five protocol families

Estimator	MAE to prefix ref.	efficiency resid.	dummy credit	Top-action match
Precedence Shapley	0.0067	0.000	0.000	100.0%
Position	0.2387	0.000	0.143	100.0%
Vanilla Shapley	0.2567	0.000	0.000	0.0%
Uniform	0.2800	0.000	0.167	0.0%
Cost	0.2830	0.000	0.172	12.5%
Leave-one-out	0.5733	3.440	0.000	0.0%

(d) Convergence to the exact reference

Uniform samples over valid linear extensions; 95% CI across 12 seeds

